Student's *t*-distribution

https://en.wikipedia.org/wiki/Student%27s_t-distribution

Student's t-distribution (or simply the t-distribution)
is any member of a family of continuous probability distributions
that arises when **estimating the mean of a normally distributed population**
in situations **where the sample size is small**
and the population standard deviation is unknown.
It was developed by **William Sealy Gosset** (1908)
under the pseudonym **Student**.

Let X_i be independently and identically drawn from the normal distribution with expected mean value **μ** and variance **σ²**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

----- Sample mean

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

----- Sample variance

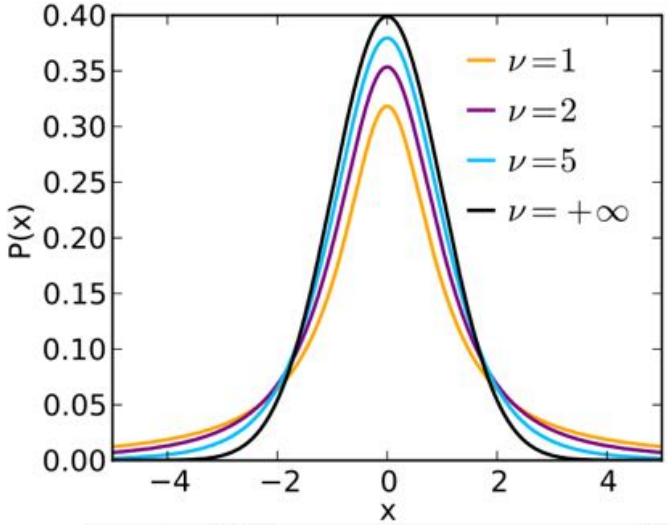
$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

----- Random variable with Normal distribution N(0,1)

$$\frac{\bar{X}-\mu}{S/\sqrt{n}};$$

----- Random variable, **t-distribution**. It uses sample variance (because real sigma is unknown). This creates **uncertainty** – and leads to the "**fat tails**" in the t-distribution.

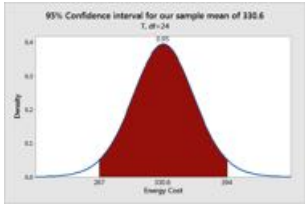
William Sealy Gosset used pseudonym "Student" because he was working for **Guinness Beer Company** in England as a "**Head Experimental Brewer**" (working on cultivating barley), and the company didn't allow him to publish under his own name.



$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

In Excel: T.DIST(x,DF,FALSE)
where DF = Degrees of Freedom (Npoints-1)

95% confidence interval



t-statistics vs z-statistics

z-statistics:

When sample size is big (> 30), then in many cases sample distribution is close to gaussian. So we use gaussian distribution and z-statistics.

Suppose that variable x follows normal gaussian distribution N(μ, σ) with mean value **μ** and standard deviation **σ**.
Then we can compose variable "**z**" which would follow standard normalized distribution N(0,1) with zero mean and standard deviation = 1 like this:

z = (x-μ) / σ

Z-score is simply the z-value which shows how far is the value from the mean expressed in number of "sigmas".

For example, if z > 2, that means that the original value is more than two sigmas away from the mean value. For normal distribution this means that the probability of this event is less than 0.05. (to be precise, the exact z-score for 0.05 is 1.96)

Suppose we take a sample of n values from N(μ, σ).
Let "**X**" (big X) be the sample mean. Then it follows normal distribution N(μ, σ/√(n)) with mean value **μ** and standard deviation **σ/√(n)** .
So the standard deviation of the mean gets smaller with larger "n".

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$ - Random variable with Normal distribution N(0,1)

t-statistics:

When sample size is small (N < 30), then even if we know that the underlying distribution is Gaussian, we still can not evaluate the distribution's sigma with enough precision. In this case we need to use t-distribution and t-statistics.

t = (X – μ) / [s/√(n)]

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Here **μ** is a sample mean, **s** is a sample standard deviation.

z-distribution (normal) gives 95% confidence at z <= 2 .

t-distribution has "fat tails", so to get 95% confidence interval we need to use t-score larger than 2.
How much larger decreases with increasing the N of points in the sample.

df = degrees of freedom = N-1

```
N-1    95% t-score
------------------
 1   : 12.71
 2   :  4.30
 3   :  3.18
 5   :  2.57
10   :  2.23
20   :  2.09
30   :  2.04 – almost normal
60   :  2.00 – almost normal
------------------
```

```
from scipy import stats
# Student, p<0.05, 2-tail, same as Excel TINV(0.05,n-1)
for df in [1,2,3,5,10,20,30,60]:
    t_score = stats.t.ppf(1-0.025, df)
    print(f"{df:2}   : {t_score:5.2f}")
```

Hypothesis Tests with the *t* Statistic

The one-sample t test from: https://www.statisticshowto.com/one-sample-t-test/

Your company wants to improve sales. Past sales data indicate that the average sale was $100 per transaction. After training your sales force, recent sales data (taken from a sample of 25 salesmen) indicates an average sale of $130, with a standard deviation of $15.
Did the training work? Test your hypothesis at a 5% alpha level.

Step 1: Write your null hypothesis. H0: μ = $100 , has not changed.
Step 2: Write your alternate hypothesis. H1: μ > $100 , increased.
Step 3: Calculate some items:
The sample mean($\bar{x}$). This is given in the question as $130.
The population mean(μ). Given as $100 (from past data).
The sample standard deviation(s) = $15.
Number of observations(n) = 25.
Step 4: Insert the items from above into the t score formula.

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

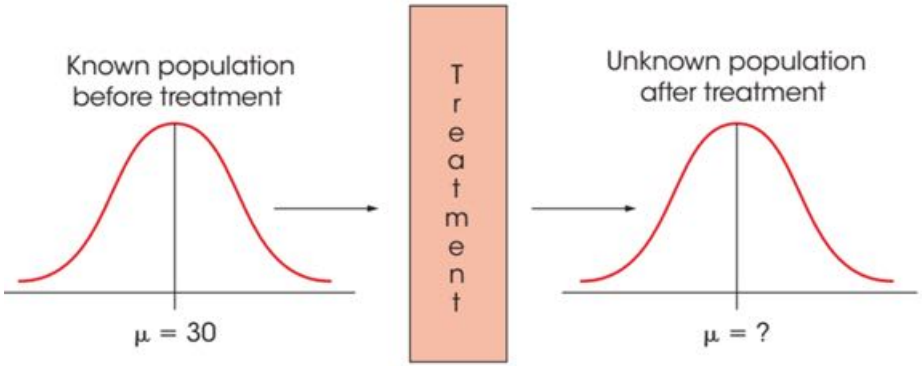
t = (130 – 100) / ((15 / √(25)) = 30/3 = 10

Step 5: Find the t-table value for alpha = 5%
and degrees of freedom = Number of observations – 1 = 24.
The intersection is 1.711 . This is your one-tailed critical t-value.
As 10 > 1.711, we reject the null hypothesis. it's highly likely that the mean sale has become greater. The sales training was probably a success.

Three types of t-tests:

- **one-sample t-test** - compare sample mean with given value
- **two-sample t-test** (a.k.a. Independent Samples t Test) compares the means of two independent groups
- **paired t-test** (for dependent samples - before/after treatment)

Common Basic Research Situation for t Statistic:



Assumed that μ is known for the population before treatment.
The purpose of the research study is to determine whether the treatment has an effect.
Note that the population after treatment has unknown values for the mean and the variance. So we will use a sample to test a hypothesis about the population mean.

Independent t-test formula

Student t-test formula

If the variance of the two groups are equivalent (**homoscedasticity**), the t-test value, comparing the two samples (A and B), can be calculated as following:

$$t = \frac{m_A - m_B}{\sqrt{\frac{S^2}{n_A} + \frac{S^2}{n_B}}}$$

where,

- m_A and m_B represent the mean value of the group A and B, respectively.
- n_A and n_B represent the sizes of the group A and B, respectively.
- S^2 is an estimator of the pooled variance of the two groups.
  It can be calculated as following :

$$S^2 = \frac{\sum (x - m_A)^2 + \sum (x - m_B)^2}{n_A + n_B - 2}$$

Degrees of freedom (df): df=nA+nB−2

Paired t-test formula

The **paired t-test** is used to compare the means of two related groups of samples

$$t = \frac{m}{s/\sqrt{n}}$$

where,

- m is the mean differences
- n is the sample size (i.e., size of d)
- s is the standard deviation of d
- Degrees of freedom (df): df=n−1

When the two independent samples are assumed to be drawn from populations with **unequal variances**,
the test statistic t is computed as **Welch's t test**:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

n_1, n_2 = sample sizes (i.e., number of observations)

$\bar{x}_1$, $\bar{x}_2$ = mean values of samples

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

s_1 , s_2 = standard deviations of samples

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

The degrees of freedom of Welch t-test is estimated as following:

$$df = \left(\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}\right)^2 / \left(\frac{S_A^4}{n_A^2(n_A - 1)} + \frac{S_B^4}{n_B^2(n_B - 1)}\right)$$

t-test – continued

Effect size.

In addition to rejecting/accepting a Hypothesis, you can also estimate the size of the effect:

- **Cohen's d** - for comparing two samples/groups with same sizes and similar SDs.
- **Glass's delta** - uses only the SD of the control group.
- **Hedges' g** - for two groups with different sample sizes.

Example: for two groups with mean M and standard deviations SD :

Cohen's d = $(M_2 - M_1)/SD_{pooled}$

where

$SD_{pooled} = \sqrt{((SD_1^2 + SD_2^2)/2)}$

Determining the amount of variability in scores explained by the treatment effect is an alternative method for measuring effect size.

$$r^2 = \frac{\text{variability accounted for}}{\text{total variability}} = \frac{t^2}{t^2 + df}$$

$r^2 = 0.01$ small effect
$r^2 = 0.09$ medium effect
$r^2 = 0.25$ large effect

Estimating "μ" from sample

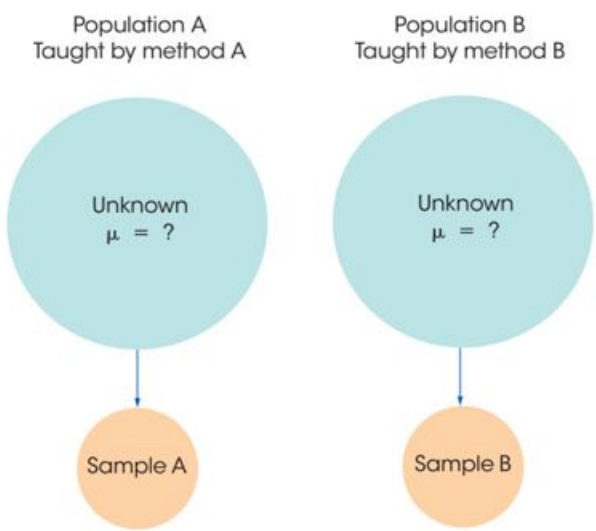
(use sample mean "M" and sample deviation S_M)

$$t = \frac{M - \mu}{s_M} \quad \Rightarrow \quad \mu = M \pm t s_M$$

For any α we know that (1 – α) proportion of t values fall between ± t for the appropriate df.

The t Test for Two Independent Samples

Data comes from two completely different, independent groups.
Observations within each sample must be independent.
The two populations must be normal.
The two populations must have equal variances (called homogeneity of variance).



Null hypothesis for independent-measures test:

$$H_0 : \mu_1 - \mu_2 = 0$$

Basic structure of the t statistic

$$t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{s_{(M_1 - M_2)}}$$

where estimated standard error (unbiased if n1=n2)

$$s_{(M_1 - M_2)} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

unbiased estimate:

$$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$$

SS = Sum of Squares
Degrees of freedom df = $df_1 + df_2 = (n_1 - 1) + (n_2 - 1)$

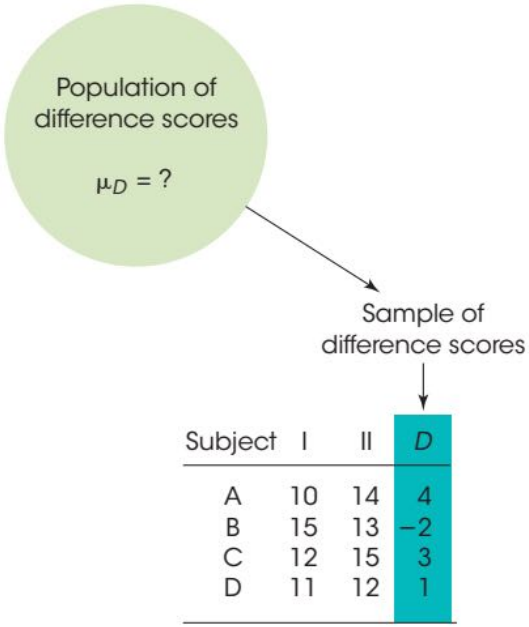
Test for homogeneity of variance:
Levene's test or Brown–Forsythe test
https://en.wikipedia.org/wiki/Levene%27s_test

The t Test for Repeated-Measures

Repeated-measures design
(a.k.a. within-subjects design)

Two separate scores are obtained **for each individual in the sample**

Same subjects are used in both groups.
Observations within each treatment condition must be independent.
Population distribution of difference scores (D values) must be normally distributed.



Subject	I	II	D
A	10	14	4
B	15	13	−2
C	12	15	3
D	11	12	1

Difference score = $D = X_2 - X_1$

Mean Difference: $M_D = \frac{\sum D}{n}$

Null Hypothesis H_0: $\mu_D = 0$

$$t = \frac{M_D - \mu_D}{s_{M_D}}$$

$$\text{variance of D} = s^2 = \frac{SS}{df}$$

$$s_{M_D} = \sqrt{\frac{s^2}{n}}$$

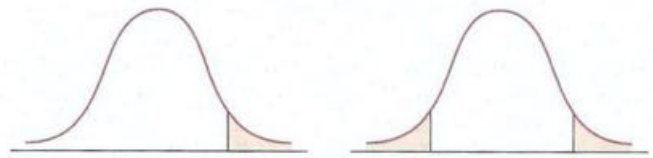
$$estimated\ Cohen's\ d = \frac{M_D}{s}$$

$$r^2 = \frac{t^2}{t^2 + df}$$

Confidence Interval (C.I.):

$$C.I.(1-\alpha): \mu_D = M_D \pm t_\alpha s_{M_D}$$

t



One tail (either right or left) — Two tails combined

	Proportion in One Tail					
	0.25	0.10	0.05	0.025	0.01	0.005
	Proportion in Two Tails Combined					
df	0.50	0.20	0.10	0.05	0.02	0.01
1	1.000	3.078	6.314	12.706	31.821	63.657
2	0.816	1.886	2.920	4.303	6.965	9.925
3	0.765	1.638	2.353	3.182	4.541	5.841
4	0.741	1.533	2.132	2.776	3.747	4.604
5	0.727	1.476	2.015	2.571	3.365	4.032
6	0.718	1.440	1.943	2.447	3.143	3.707
7	0.711	1.415	1.895	2.365	2.998	3.499
8	0.706	1.397	1.860	2.306	2.896	3.355
9	0.703	1.383	1.833	2.262	2.821	3.250
10	0.700	1.372	1.812	2.228	2.764	3.169
11	0.697	1.363	1.796	2.201	2.718	3.106
12	0.695	1.356	1.782	2.179	2.681	3.055
13	0.694	1.350	1.771	2.160	2.650	3.012
14	0.692	1.345	1.761	2.145	2.624	2.977
15	0.691	1.341	1.753	2.131	2.602	2.947
16	0.690	1.337	1.746	2.120	2.583	2.921
17	0.689	1.333	1.740	2.110	2.567	2.898
18	0.688	1.330	1.734	2.101	2.552	2.878
19	0.688	1.328	1.729	2.093	2.539	2.861
20	0.687	1.325	1.725	2.086	2.528	2.845
21	0.686	1.323	1.721	2.080	2.518	2.831
22	0.686	1.321	1.717	2.074	2.508	2.819
23	0.685	1.319	1.714	2.069	2.500	2.807
24	0.685	1.318	1.711	2.064	2.492	2.797
25	0.684	1.316	1.708	2.060	2.485	2.787
26	0.684	1.315	1.706	2.056	2.479	2.779
27	0.684	1.314	1.703	2.052	2.473	2.771
28	0.683	1.313	1.701	2.048	2.467	2.763
29	0.683	1.311	1.699	2.045	2.462	2.756
30	0.683	1.310	1.697	2.042	2.457	2.750
40	0.681	1.303	1.684	2.021	2.423	2.704
60	0.679	1.296	1.671	2.000	2.390	2.660
120	0.677	1.289	1.658	1.980	2.358	2.617
∞	0.674	1.282	1.645	1.960	2.326	2.576